# Recording L01-R01
# Gaussian conditioning by block elimination

A normalized density proof of the Gaussian conditioning formulas

Probabilistic Machine Learning 2027-1

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# From a joint Gaussian to a conditional density

**From the lecture** L01 live lecture: conditioning mean and covariance; independent residual

$$m(x_2) = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

$$S = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

**To prove here**

$$p(x_1, x_2) = \mathcal{N}(x|\mu, \Sigma)$$

$$p(x_1|x_2) = \mathcal{N}(x_1|m(x_2), S)$$

Assume $\Sigma = \Sigma^T \succ 0$ is symmetric positive definite: $v^T\Sigma v > 0$ for $v \neq 0$. Here $x_i, \mu_i \in \mathbb{R}^{d_i}$ and $d = d_1 + d_2$.

# The joint density has a quadratic and a normalizing factor

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$p(x_1, x_2) = \frac{\exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right]}{(2\pi)^{\frac{d}{2}}\sqrt{\det \Sigma}}$$

$$Q = (x-\mu)^T \Sigma^{-1}(x-\mu), \quad C = \frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{\det \Sigma}}$$

$$p(x_1, x_2) = C \exp\left(-\frac{Q}{2}\right)$$

> $Q$ controls the exponential; $C$ is the factor that makes the density integrate to one. A conditional density requires both parts, with their correct constants.

# The observed covariance block is invertible

**Premise** Orientation

$$\Sigma = \Sigma^T \succ 0$$

$$z \in \mathbb{R}^{d_2}, \quad z \neq 0, \quad \begin{pmatrix} 0 \\ z \end{pmatrix} \neq 0$$

$$z^T \Sigma_{22} z = \begin{pmatrix} 0 \\ z \end{pmatrix}^T \Sigma \begin{pmatrix} 0 \\ z \end{pmatrix} > 0$$

$$\Sigma_{22} = \Sigma_{22}^T \succ 0 \quad \Rightarrow \quad \Sigma_{22}^{-1} \text{ exists}$$

Positive definiteness of the joint covariance also holds on the observed coordinate subspace.

# Choose the row subtraction to cancel the cross block

**First block row minus $B$ times the second**

$$T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}, \quad T\Sigma = \begin{pmatrix} \Sigma_{11} - B\Sigma_{21} & \Sigma_{12} - \mathrm{B}\Sigma_{22} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

**Choose $B$ to make the upper-right block zero**

$$\Sigma_{12} - B\Sigma_{22} = 0$$

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$T\Sigma = \begin{pmatrix} \Sigma_{11} - B\Sigma_{21} & 0 \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

A block diagonal covariance has a block diagonal inverse. In those transformed coordinates, the quadratic separates into two block contributions.

# The matching column subtraction leaves a block diagonal matrix

**First block column minus the second times $B^T$**

$$T\Sigma T^T = \begin{pmatrix} \Sigma_{11} - B\Sigma_{21} & 0 \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} I & 0 \\ -B^T & I \end{pmatrix}$$

$$\Sigma_{22} B^T = \Sigma_{21}$$

$$T\Sigma T^T = \begin{pmatrix} \Sigma_{11} - B\Sigma_{21} & 0 \\ \underbrace{\Sigma_{21} - \Sigma_{22} B^T}_{=0} & \Sigma_{22} \end{pmatrix} = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

$$S = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

The remaining first block is the Schur complement of $\Sigma_{22}$ in $\Sigma$, the same residual covariance obtained in the live lecture.

# The remaining covariance block is positive definite

**Elimination** Previous slide

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}, \quad T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}$$

$$a \in \mathbb{R}^{d_1}, \quad a \neq 0, \quad T^T \begin{pmatrix} a \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ -B^T a \end{pmatrix} \neq 0$$

$$a^T S a = \begin{pmatrix} a \\ 0 \end{pmatrix}^T T\Sigma T^T \begin{pmatrix} a \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} a \\ -B^T a \end{pmatrix}^T \Sigma \begin{pmatrix} a \\ -B^T a \end{pmatrix} > 0$$

$$S = S^T \succ 0 \quad \Rightarrow \quad S^{-1} \text{ exists}, \quad \det S > 0$$

> The residual covariance is a valid nondegenerate Gaussian covariance. Both covariance blocks now have inverses and positive determinants.

# The same subtraction produces the centered residual

$$u = x_2 - \mu_2, \quad r = x_1 - \mu_1 - Bu = x_1 - m(x_2)$$

$$T(x-\mu) = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} x_1 - \mu_1 \\ u \end{pmatrix} = \begin{pmatrix} r \\ u \end{pmatrix}$$

$$T^{-1} = \begin{pmatrix} I & B \\ 0 & I \end{pmatrix}, \quad T^{-1} \begin{pmatrix} r \\ u \end{pmatrix} = \begin{pmatrix} r + Bu \\ u \end{pmatrix} = x - \mu$$

$$TT^{-1} = \begin{pmatrix} I & B - B \\ 0 & I \end{pmatrix} = I = T^{-1}T$$

> $u$ is fixed when conditioning; $r$ is the remaining centered variation around the observation-dependent mean.

# The eliminated covariance gives the inverse in the density

**Covariance identity** Slide 6; $T$ is invertible and both diagonal blocks are positive definite

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

**Invert both sides; reverse the order of factors**

$$(T^T)^{-1}\Sigma^{-1}T^{-1} = \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix}$$

**Multiply on the left by $T^T$ and on the right by $T$**

$$\Sigma^{-1} = T^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} T$$

The recovered inverse covariance is the matrix in $Q$. Its factor order lets the subtraction act directly on the centered data vector.

# Use the eliminated inverse in the quadratic

**Inputs** Joint density and previous slide

$$Q = (x-\mu)^T \Sigma^{-1} (x-\mu)$$

$$\Sigma^{-1} = T^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} T$$

**Substitute the inverse;** $(Tv)^T = v^T T^T$

$$Q = [T(x-\mu)]^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} [T(x-\mu)]$$

> The outer factors $T$ act on the centered vector; the middle matrix is block diagonal.

# Block diagonal multiplication separates the quadratic

**Inputs** Previous slide and residual coordinates

$$Q = [T(x-\mu)]^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} [T(x-\mu)]$$

$$T(x-\mu) = \begin{pmatrix} r \\ u \end{pmatrix}, \quad r = x_1 - m(x_2), \quad u = x_2 - \mu_2$$

$$Q = \begin{pmatrix} r \\ u \end{pmatrix}^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} r \\ u \end{pmatrix}$$

$$= \underbrace{r^T S^{-1} r}_{\text{residual}} + \underbrace{u^T \Sigma_{22}^{-1} u}_{\text{observed deviation}}$$

For fixed $x_2$, the observed term is constant in $x_1$; the residual term varies through $r = x_1 - m(x_2)$.

# The same elimination also splits the determinant

**Covariance identity** Elimination

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}, \quad T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}$$

$$\det T = \det I_{d_1} \det I_{d_2} = 1$$

**$T$ is triangular, so its determinant is $1$; determinants multiply**

$$\det\left(T\Sigma T^T\right) = \det T \cdot \det \Sigma \cdot \det T^T = \det \Sigma$$

$$\det \Sigma = \det S \cdot \det \Sigma_{22}$$

> The elimination preserves the determinant and exposes exactly the two covariance determinants needed for Gaussian normalization.

# Dimensions and determinants split the normalizing factor

**Coefficient** Full density and block dimensions

$$C = (2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}}, \quad d = d_1 + d_2$$

$$C = (2\pi)^{-\frac{d_1+d_2}{2}} (\det S \cdot \det \Sigma_{22})^{-\frac{1}{2}}$$

**Split the powers and positive square roots**

$$C = \underbrace{\frac{1}{(2\pi)^{\frac{d_1}{2}} \sqrt{\det S}}}_{\text{residual factor}} \cdot \underbrace{\frac{1}{(2\pi)^{\frac{d_2}{2}} \sqrt{\det \Sigma_{22}}}}_{\text{observed factor}}$$

Each coefficient has the dimension and covariance determinant of its Gaussian density; positivity licenses the positive square roots.

# Both parts combine into normalized Gaussian factors

**Quadratic** Quadratic split

$$Q = r^T S^{-1} r + u^T \Sigma_{22}^{-1} u, \quad u = x_2 - \mu_2, \quad r = x_1 - m(x_2)$$

**Substitute the separated quadratic and coefficient in** $C \exp\left(-\frac{Q}{2}\right)$

$$p(x_1, x_2) = \frac{\exp\left[-\frac{1}{2} r^T S^{-1} r\right]}{(2\pi)^{\frac{d_1}{2}} \sqrt{\det S}} \cdot \frac{\exp\left[-\frac{1}{2} u^T \Sigma_{22}^{-1} u\right]}{(2\pi)^{\frac{d_2}{2}} \sqrt{\det \Sigma_{22}}}$$

$$p(x_1, x_2) = \mathcal{N}(x_1 | m(x_2), S) \mathcal{N}(x_2 | \mu_2, \Sigma_{22})$$

For each fixed $x_2$, the first factor is a normalized density in $x_1$. Its mean still depends on the observation.

# Integrating out the residual identifies the marginal

**Joint density** Previous slide

$$p(x_1, x_2) = \mathcal{N}(x_1 | m(x_2), S)\mathcal{N}(x_2 | \mu_2, \Sigma_{22})$$

**Hold $x_2$ fixed and integrate with respect to $x_1$**

$$p(x_2) = \int_{\mathbb{R}^{d_1}} p(x_1, x_2)\, \mathrm{d}x_1$$

$$= \mathcal{N}(x_2 | \mu_2, \Sigma_{22}) \underbrace{\int_{\mathbb{R}^{d_1}} \mathcal{N}(x_1 | m(x_2), S)\, \mathrm{d}x_1}_{=1}$$

$$p(x_2) = \mathcal{N}(x_2 | \mu_2, \Sigma_{22}) > 0$$

The factor that depends only on the observation is its marginal density. Its strict positivity makes the density ratio available at every $x_2$.

# Dividing by the marginal gives the conditional Gaussian

**Cancel the positive observed-block density**

$$p(x_1|x_2) = \frac{p(x_1,x_2)}{p(x_2)} = \frac{\mathcal{N}(x_1|m(x_2),S)\mathcal{N}(x_2|\mu_2,\Sigma_{22})}{\mathcal{N}(x_2|\mu_2,\Sigma_{22})} = \mathcal{N}(x_1|m(x_2),S)$$

$$m(x_2) = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

$$S = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

> The observed deviation shifts the conditional mean. The conditional covariance is the residual covariance and does not depend on the observed value.

# Zero cross-covariance removes the observation

$$\Sigma_{12} = 0 \quad \Rightarrow \quad B = 0, \quad m(x_2) = \mu_1, \quad S = \Sigma_{11}$$

$$p(x_1|x_2) = \mathcal{N}(x_1|\mu_1, \Sigma_{11}) = p(x_1)$$

$$p(x_1, x_2) = p(x_1)p(x_2) \quad \Rightarrow \quad X_1 \perp X_2$$

For a joint Gaussian, zero cross-covariance makes observations of one block uninformative about the other.